OXFORD AQA
INTERNATIONAL QUALIFICATIONS

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International AS and A-level

Biology (9610)
BL05 Synoptic paper
Report on the examination

June 2024

REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL BIOLOGY (9610) BL05 JUNE 2024

General

There were some good answers to each question where students demonstrated a high level of understanding of material from the Specification. The most successful answers were for the calculation of a magnification, the refinement of a practical technique to find the optimum temperature for activity of an enzyme, the understanding of spatial summation for enhancing the sensitivity of the eye to light and the use of a high temperature for separating the strands of DNA.

Areas causing some difficulty were drawing from a photograph, the selection of relevant material for an answer, and striking an appropriate balance between depth and breadth of coverage in an extended prose answer.

Students should be encouraged to read carefully through information given in the stem of a question as this is provided specifically for their guidance and sets the scene for making a reasoned response: time would be well spent on assimilating the information provided and in making sure it is well understood before attempting to answer any questions based upon it. However, no marks are available for simply quoting such information.

It is also vital to take note of any particular requirement of a question, for example, in the presentation of data in a calculation, where an answer requires either a percentage (rather than a proportion) or a specified number of significant figures.

QUESTION 01

This question was about the structure of xylem tissue, transpiration, and the experimental measurement of water uptake using a potometer.

01.1

As in previous examinations, drawings of the five xylem vessels in **Figure 1** were generally weak. Many ignored the instruction to 'include the connection of the xylem vessels to the adjacent cells' and either drew *all* the adjacent cells or included *no* connections. Some drawings were entirely imaginary, including 3-dimensional representations of xylem vessels. Many students found difficulty in maintaining the shapes and proportions given in the photograph. Only about 1 in 12 scored full marks.

01.2

Most students used the sides of the rectangles in **Figures 1** and **2** as the basis for their calculation of the magnification of **Figure 2**, ie either the heights (68 and 32 mm) or the widths (53 and 25 mm) to calculate a magnification of $\times$1060. A tolerance of $\pm$ 1 mm was allowed on the measurements with appropriate allowance being made for the final answer. Some students measured part of their drawing in **Figure 2** and thus used different numbers, which was permissible provided they had shown what they had done; unfortunately, many did not do this and thus their numbers did not convey any meaning. A few students obtained answers to an incorrect order of magnitude (eg $\times$106) or did not follow the instruction to give the answer to 3 significant figures (eg $\times$1062.5). More than half the students scored full marks for their calculation.

01.3

The most common point made in this answer was an appropriate reference to the cohesion of water molecules via hydrogen bonds to enable water movement up the xylem. Many explained that transpiration occurred mainly via the stomata but evaporation, or the loss of water vapour, was referred to less often. Better students explained how a reduction in water potential in the leaf cells would result in water moving from the xylem vessels from a higher to a lower water potential (movement down a ‘concentration’ gradient was ignored by examiners). Many students included irrelevant details such as root pressure or the movement of water across the tissues of the root via the apoplastic and symplastic pathways. The question differentiated well across the ability range.

01.4

This question required a description of how the potometer shown in **Figure 3** could be used for estimating the rate of transpiration. It related to Required Practical Activity 6. It did *not* require a description of how the apparatus should have been set up, although a large proportion of students devoted the bulk of their answer to this. Most students described the measurement of the distance moved by the air bubble over a set time and how the bubble could be returned to the lower end of the millimetre scale using the syringe. An appropriate number of repetitions and calculation of a mean was also included by many, but suitable control variables, such as air temperature (the temperature of water in the beaker was *not* regarded as a suitable control variable) or air movements or light intensity, were rarely mentioned. Better students explained how the required rate (in mm^3 per hour) could be calculated using $\pi r^2 h$ and adjusting for the time taken. This question differentiated well and over half scored above half-marks.

01.5

The most common suggestion of why the potometer gave only an approximate value for the rate of transpiration was that some of the water taken in by the plant would be used in photosynthesis. Other uses, such as hydrolysis or the maintenance of turgor, were suggested by some students. There were only occasional references to the fact that the apparatus actually measured water uptake rather than water loss.

QUESTION 02

This question was about the experimental determination of the optimum temperature for the activity of the enzyme amylase. It related to Required Practical Activity 1.

02.1

Students were generally successful at identifying errors in the given method, the most common being failure to incubate the starch at the temperature being tested before mixing with the amylase, the addition of a ‘few drops’ of amylase solution rather than a measured volume and that the pH had not been controlled by using a buffer solution. Additional errors identified by some were the long 2-minute intervals for testing with iodine solution, not stirring / shaking the reaction mixture as soon as the amylase had been added and the large 20°C intervals between the temperatures being tested.

02.2

The most common errors identified in the graph of results were the large circles used for plotting the points instead of the use of crosses, and the scale on the y-axis being too small – some recommending that it should be doubled and a few making the observation that the spread of date should have covered more than half of the available axis. Relatively few noted that units had been omitted from the axes; some recommended that a smooth curve of best fit would have been better than joining the points with a series of ruled lines. Hardly any commented on the inappropriate extrapolation to 0°C but many thought that error bars should have been included – even though the experiment had been conducted only once. More than three-quarters of students were able to suggest at least one improvement.

02.3

Most students knew that testing more temperatures at closer intervals around the supposed optimum of 40°C would have helped to obtain a more accurate value for the optimum. A common error was to specify a range of temperatures on just one side of 40°C, for example just at 25, 30, 35 and 40°C. Two-thirds of students scored full marks.

QUESTION 03

This question was about Huntington's disease and its occurrence in one isolated area in South America.

03.1

The vast majority of students understood that the genotypes that caused Huntington's were **HH** and **Hh**.

03.2

Over a third of students gained full marks in this question which involved the use of the Hardy-Weinberg equation for calculating the expected percentage of people who were heterozygous for Huntington's. The main error was thinking that the given percentage of people with the **H** allele (48%) represented the frequency of the **H** allele in the population rather than indicating that 52% did not have this allele, hence $0.52 = q^2$, $q = 0.7211$, $p = 1 – 0.7211 = 0.2789$, thus $2pq = 0.4022$ and the percentage of heterozygotes was 40%. Examiners did allow up to 2 compensation marks (out of 3) for an answer correctly derived from an incorrect initial assumption. A few students forgot to convert their decimal proportion back into a percentage.

0.33

Explanations of why the **H** allele had not been lost from this population due to natural selection were often inadequate, with only one in 25 students scoring full marks. The most common correct idea was that most people with the **H** allele would have already reproduced before they showed any signs of the condition. Many mentioned lack of gene flow but usually did not give sufficient detail in application to the given situation; better students explained that this meant there was a high probability of mating with someone who had the **H** allele.

03.4

Most students were able to give at least a partial explanation, using the given information, of how the protein huntingtin could lead to the death of neurones in the brain. Most of these mentioned a reduction in ATP synthesis but only a third also explained that reduced NAD would be unable to pass electrons to the electron transfer chain in the mitochondrion. Weaker students just thought that oxygen would not be used.

QUESTION 04

This question was about sensitivity, acuity and colour vision in the human eye.

04.1

Nearly all students correctly identified cells **W** and **Z** in the diagram of the retina as being, respectively, rods and cones.

04.2

The concept of summation within a group of rod cells leading to the ability to detect low-intensity light was well understood although some students lost marks due to poor expression – eg for stating that rods were connected to more than one bipolar neurone rather that making the correct observation that more than one rod was connected to a single bipolar neurone. The idea that a threshold level needed to be met before an action potential was generated in the neurone was generally understood.

04.3

Over half the students scored full marks for using information from the diagram that showed each cone connected to a separate bipolar neurone and hence able to send separate sets of impulses to the brain, resulting in high visual acuity.

04.4

Although many students knew that cones contained different pigments sensitive to different wavelengths / colours of light, it was not always clear in their answers that each cone contained just one of the three types of pigment. Only 1 in 5 students described this adequately. Very few students made the additional point that our appreciation of a range of colours was due to varying degrees of stimulation of the three types of cone.

QUESTION 05

This question introduced the topic of using DNA probes for detecting the allele for cystic fibrosis in a fetus and the evaluation of a gene therapy technique for treating a patient with cystic fibrosis.

05.1

More than three-quarters of students were able to apply what they had learned in the context of PCR to this new situation and explain that heating the DNA to 95°C would break hydrogen bonds and produce two single strands of DNA.

05.2

Although two-thirds understood that cooling the DNA to 50°C before mixing with the probe would allow the probe to bind to the DNA, far fewer understood that cooling to 20°C would have allowed the original long strands of DNA to anneal, which was not wanted. Many thought that the binding would 'be too slow' at 20°C and there was much inappropriate mention of varying levels of enzyme activity at the different temperatures, some even thinking that these enzymes would 'denature' at 20°C.

05.3

Satisfactory explanations of why the two different probes would bind to different alleles of the CFTR gene were given by half of the students. While complementarity was generally understood, inadequate answers failed to stress that *sequences* of bases would be complementary, with some describing how ‘tertiary structures’ would be complementary.

05.4

Two-thirds of students gave inadequate answers to this question, mainly stating that the fluorescent probes acted as ‘gene markers’ rather than explaining that they enabled the scientists to detect the different alleles.

05.5

Interpretation of the data in this question required careful reading of the information given in the stem of the question in order to understand that one probe was longer than the other, that the shorter probe would move faster than the longer probe during electrophoresis and that the longer probe would have combined with the **n** allele while the shorter probe bound to the **N** allele. Just over one-third were completely successful. An amazing number of students thought that the trace with two peaks represented a homozygous condition while a trace with one peak would come from a heterozygote.

Questions 05.6 and 05.7 concerned evaluation of the methods used by the scientists and their results for an investigation of a gene-therapy technique for treating patients with cystic fibrosis.

05.6

A little over half the students realised that a double-blind trial, with some patients receiving the CFTR allele and some a placebo, would overcome bias or psychological effects. Better students added that this would make the investigation more valid or would give reliable results. Some spoiled their answers by describing the results as being more ‘accurate.’

05.7

In their evaluation of the methods and results of the scientists’ investigation, nearly all students considered points both supporting the success of the treatment and not supporting it. The most successful answers referred to the overall improvement in patients receiving the CFTR allele compared to the placebo group and to the *significant* improvement at 30 days in the patients receiving the CFTR allele compared to those receiving the placebo as indicated by the non-overlap of the standard deviations from the two groups. This was countered by the decrease in the improvement after 30 days. The small sample size was also frequently commented upon, but this point only scored a mark if accompanied by the observation that this would reduce validity or would not be sufficiently representative. Points rarely given related to the actual measurement made (speed of breathing out at the given times compared with that at the start) possibly not being the only indication of lung function, and to the possible variation in the severity of the condition in the different patients which might have made the results non-comparable. The question differentiated well between students of differing ability with 1 in 17 scoring full marks, almost half scoring at least half marks and almost 90 percent making one or more valid observations.

QUESTION 06

This question was about plants and water and required three extended prose answers on the topics of:

- the processes in plants that use water
- the experimental determination of the water potential of a plant tissue
- the adaptations of xerophytic plants.

Each of the three sections differentiated well between students of different abilities and the mean mark for each section was around 3 out of 6 marks.

06.1

The most common aspect included in students' answers was the photolysis of water leading to the production of ATP and reduced NADP. Water as a medium for transport of organic molecules or inorganic ions and as a medium for chemical reactions were also frequently covered, as were the opening and closing of stomata and, to a lesser extent, the support functioning of water in turgor pressure. Many students also mentioned the use of water in hydrolysis reactions – eg the hydrolysis of starch or of ATP and, occasionally, cooling of the plant by evaporation of water.

Many students concentrated too much on just one or two topics at the expense of breadth of coverage – eg superfluous details of symplastic and apoplastic pathways of water transport, or perhaps devoting most of their answer to the process of photosynthesis. One in 8 students scored full marks and this question was the best discriminator on the paper.

06.2

Most students described how mass changes in a plant tissue, such as potato, could be measured due to immersion in different concentrations of sucrose solution. It was common for students to omit several of the following essential details:

- ensuring the tissue samples were the same size with the same surface area – eg by cutting a cylinder of tissue with a cork borer and then cutting to the same length
- blotting the tissue dry before weighing and immersing in the sucrose solutions
- giving at least 5 concentrations of sucrose with stated units – eg between 0 and 1.0 mol dm^{-3}
- leaving the tissue in the solutions for a suitable time (20 minutes was considered a minimum)
- blotting the tissue dry again before re-weighing
- giving a suitable number of repetitions at each concentration (at least twice more)
- calculating a mean percentage change in mass for each concentration
- plotting a graph and drawing a smooth curve to determine the concentration giving zero change in mass
- conversion of the isotonic concentration into a water potential using published data.

A common error was the addition of a buffer solution which, of course, would have altered the water potential of each solution.

One in 7 students scored full marks in this section and almost half scored at least 4 out of the 6 marks available.

06.3

The most common adaptation of xerophytes given by students was the extensive root system – either penetrating deep into the soil or spreading over a wide area near the soil surface. Having small, narrow leaves or leaves that could be rolled up to minimise the surface area exposed to the air and having sunken stomata or stomata that opened at night, again to minimise exposure or to open only when the surrounding air was cooler, were features mentioned by many. Many students mentioned the waxy cuticle but many of these did not emphasise that the cuticle was thicker in xerophytes than in non-xerophytes or did not explain that the thick cuticle reduced / prevented the passage of water through it. Some students mentioned surface hairs on the stems and leaves which maintained a humid atmosphere near the surface and thus reduced the water potential gradient between the plant and the surrounding air. Other aspects, given by relatively few students were ridged stems for casting shade, water storage tissues and reduced heights (and hence exposure) found in some xerophytes. More than a quarter of students scored at least 4 of the 6 marks available but only 1 in 45 scored full marks.